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A prompt framework for enhancing LLM-based explainability of medical machine learning models: an intensive care unit application

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Abstract

Background Explainable AI (XAI) techniques like SHAP provide valuable insights into machine learning model predictions by quantifying feature contributions. However, interpreting these quantitative outputs remains unintuitive for many clinicians, hindering their practical adoption in clinical decision-making. This exploratory feasibility study aims to propose and evaluate a prompting framework designed to guide large language models (LLMs) in generating consistent, clinically relevant explanations from SHAP values.

Methods We developed a structured zero-shot prompting framework incorporating variable definitions, safety principles, and a three-step reasoning process; (1) key risk factors, (2) prediction-outcome reconciliation, (3) clinical recommendations. This framework guided GPT-4 to explain SHAP-based predictions from an ICU extubation failure model (trained on MIMIC-III data). The framework's performance was assessed using quantitative consistency metrics (entropy, accuracy) and a qualitative clinician survey ($n=7$).

Results The LLM achieved a fidelity of 0.783 and entropy of 0.226, indicating consistent structured reasoning. On a 5-point Likert scale, clinicians rated the LLM-generated explanations as helpful (mean: 3.94 ± 0.48) and safe (4.21 ± 0.68). However, critical care specialists assigned lower scores than non-critical care physicians (helpfulness: 3.48 vs. 4.29; safety: 3.45 vs. 4.79), suggesting domain-specific caution in perceived utility.

Conclusions A structured prompting framework shows feasibility in leveraging LLMs to enhance the clinical interpretability of SHAP explanations. This methodological approach warrants further investigation and refinement for broader clinical application.

Keywords Explainable AI, SHAP, Large language models, Extubation failure, MIMIC-III, Clinical decision support

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Introduction

Machine learning (ML) models are increasingly applied in healthcare for diagnosis, risk prediction, and treatment recommendations [1, 2]. Yet their complexity often makes them difficult to understand, limiting integration into routine clinical workflows, especially in, high-stakes decision-making environments [3]. This lack of transparency, frequently referred to as the 'black-box' problem, creates significant challenges in interpretability [3, 4]. Clinicians often find it difficult to trust or fully understand ML model outputs, especially when explanations are limited to numerical feature importance score or abstract visualizations [5]. Enhancing the explainability of these models is therefore crucial for improving clinical interpretability and fostering trust.

To address these challenges, the field of explainable AI (XAI) has emerged, developing techniques aimed at clarifying model behavior [4]. SHapley Additive exPlanations (SHAP) has become widely used [6]. SHAP provides a theoretically grounded, model-agnostic method for attributing prediction contributions to individual input features [6], assigning each input variable a contribution score, indicating how much that variable pushes the prediction toward or away from a given outcome. However, the practical utility of raw SHAP values remains limited. Clinicians often find these outputs too abstract, lacking the patient-specific context and feature interactions needed for safe decision making [5, 7–9].

Recent reviews echo these limitations. For instance, a 2024 systematic analysis revealed that common post-hoc XAI methods tend to oversimplify complex model behavior, potentially leading to inaccurate or incomplete explanations that fail to capture crucial clinical nuance [10]. Critically, the review also noted that 83% of the examined studies lacked meaningful clinician involvement, raising serious questions about the clinical validity and practical utility of much XAI research [10]. Such findings point to an "XAI-to-reality gap", where theoretical promise of explainability struggles to translate into tangible clinical value. Furthermore, the effectiveness of explanations can be paradoxical; the type and quality heavily influence whether they build or erode clinician confidence, with overly complex or irrelevant explanations potentially decreasing trust [11].

Therefore, the challenge extends beyond merely providing any explanation. It lies in effectively communicating the complex reasoning of AI models to clinical end-users, many of whom lack deep technical expertise but require sufficient understanding for safe and effective practice [12]. This creates an urgent need for explanation methods that can convert complex, quantitative XAI outputs, like SHAP values into intuitive, contextualized, and reliable narrative forms that naturally align with clinical workflows and decision-making processes [7].

Large language models (LLMs) such as GPT-4 [13] presents a compelling solution to this challenge. With their advanced natural language processing and reasoning capabilities, LLMs could serve as translators, transforming technical XAI outputs into more intuitive, narrative explanations tailored for clinical understanding [14]. While LLMs have demonstrated value in clinical documentation and patient communication [15], their dedicated use in enhancing the interpretability of predictive models for clinicians remains relatively underexplored and merits focused investigation.

We hypothesized that LLMs guided by structured prompts can function as effective clinical "interpreters", translating SHAP-based model outputs into clinician-friendly narrative explanations and thereby bridging the gap between quantitative XAI results and the contextual needs of practitioners. Importantly, this work represents an exploratory feasibility study, designed to provide preliminary evidence rather than conclusive clinical validation. To test this hypothesis and explore the feasibility of this approach, we pursued the following objectives (i) to design and implement a novel structured prompting framework intended to guide LLMs in generating clinically relevant narrative explanations from SHAP values. (ii) to evaluate the feasibility and technical characteristic (e.g., consistency, fidelity) of the explanations generated by this framework using a representative clinical case study (ICU extubation failure prediction) as a testbed. (iii) To assess preliminary clinician perceptions regarding the utility and safety of the explanations produced by our framework.

Methods

This methodological study centres on evaluating a large language model (LLM)–based explanation framework. A pre-trained XGBoost classifier was used only to generate SHAP values that seed the framework; detailed performance metrics are provided in Supplement Table S3 for completeness only, not for model benchmarking.

Study setting and input generation for framework evaluation

This study utilized data from MIMIC-III (Medical Information Mart for Intensive Care III) v1.4, a critical-care database containing 53,423 ICU stays at Beth Israel Deaconess Medical Center (2001-2012) [16]. The specific clinical context chosen to evaluate our proposed explanation framework was the prediction of extubation failure in the ICU. The study cohort included adult patients (age ≥ 18 years) who received invasive ventilation for over 24 h and had at least one documented extubation attempt. Extubation failure was defined as re-intubation within 48 h of extubation [17]. Application of these criteria resulted in a final cohort of 747 extubation events(Supplementary Table S1).

To generate explanatory inputs, SHAP values were derived from 16 clinically relevant pre-extubation features. The feature list, preprocessing procedures, and model performance are reported in the Supplementary Methods (Supplementary Tables S2-S3). For evaluation, we curated 40 representative test cases, equally distributed across true positive, true negative, false positive, and false negatives. This ensured that the framework could be assessed across different prediction-outcome scenarios. In this study design, actual outcomes ("ground truth") served as the reference standard, acknowledging that future applications could instead compare model predictions with contemporaneous clinical judgement.

The structured prompting framework

The core methodological contribution of this study is a structured prompting framework designed to guide LLMs in interpreting SHAP values and generating clinically relevant narrative explanations. This framework was developed iteratively based on qualitative analysis of preliminary LLM outputs to optimize for clarity, clinical relevance, and safety.

Early prototype often produced hallucinated or inconsistent content when raw SHAP tables were provided. To address this, we iteratively refined the prompts with rules requiring explicit citation of numerical values and error messages when data were missing. Each version was tested by generating ten completions; no fabricated values were observed in the final version.

The final framework, implemented as a zero-shot prompt for the LLM, consists of four key components: (a) Clinical persona and Variable definitions: The prompt instructs the LLM to adopt the persona of a critical care physician and provides plain-language definitions for all 16 input variables used by the baseline model. This grounds the LLM's reasoning in the relevant clinical context.; (b) Safety and Generation Principles: Specific principles were included to guide safe and responsible generation, addressing potential issue like hallucination, bias (e.g., gender bias was instructed against unless identified as an important variable by SHAP), instructing the LLM on how to handle missing input values (e.g., explicitly stating data is unavailable rather than imputing or generating arbitrary values), and ensuring generated outputs acknowledge data limitations and maintain plausibility.; (c) Three-Step Reasoning Instruction: The prompt explicitly instructs the LLM to follow a structured, three-step clinical reasoning process when generating the narrative explanation: (Step 1) Extract Key Risk Factors: The LLM receives the pre-computed SHAP vector and must sorts by absolute SHAP value. It then extract the top three features are considered the key risk factors influencing the predicted risk of extubation failure.;(Step 2) Reconcile Prediction and Outcome: compare the model's prediction with the actual observed outcome (for the test cases) and attempt to reconcile any discrepancies based on the SHAP analysis and patient data; (Step 3) Propose Clinical Plan: Based on the risk assessment and reconciliation, suggest potential next steps or monitoring considerations relevant to the patient's situation.; (d) Structured Input/Output Template: A markdown-style template was defined for both the input provided to the LLM (patient data, SHAP values) and the expected output format (the three-step narrative) to ensure standardized and easily parsable responses.

This framework was designed to elicit structured clinical reasoning from the LLM without requiring model fine-tuning or few-shot examples. In this context, zero-shot prompting refers to instructing the LLM to perform the task solely on the basis of these structured instructions, without the use of additional training or exemplar demonstrations. A simplified illustration of the prompt structure is providing in Fig. 1.

LLM implementation and narrative explanation generation

We utilized the ChatGPT interface with the GPT-4 model (May 2024 version) to generate explanations based on our prompting framework. For each of the 40 selected test cases, the structured prompt containing the patient data, SHAP values, and instructions was provided to GPT-4. The LLM was queried 10 times per case to assess the consistency of its outputs. Each generated response included a visualization (SHAP decision plot), a table of all features with their SHAP values, and the structured three-part narrative explanation corresponding to the reasoning steps defined in the framework.

Framework evaluation methods

The performance, consistency, and perceived utility of the explanations generated by our prompting framework were assessed using both quantitative and qualitative methods.

Quantitative assessment: explanation consistency and fidelity

LLM performance was assessed by evaluating the top three SHAP ranked risk factors across generations. Note that the LLM does not compute SHAP; it consumes the pre-computed SHAP vector and is required to extract the top three features by absolute value. Performance was quantified using: (1) Exact match accuracy (score = 1 if all three matched), (2) Fidelity (scored as 1/0.66/0.33 for 3/2/1 correct features), and (3) Entropy, defined as the degree of variation in SHAP triplet predictions across 10 outputs for each case. The Entropy was calculated as:

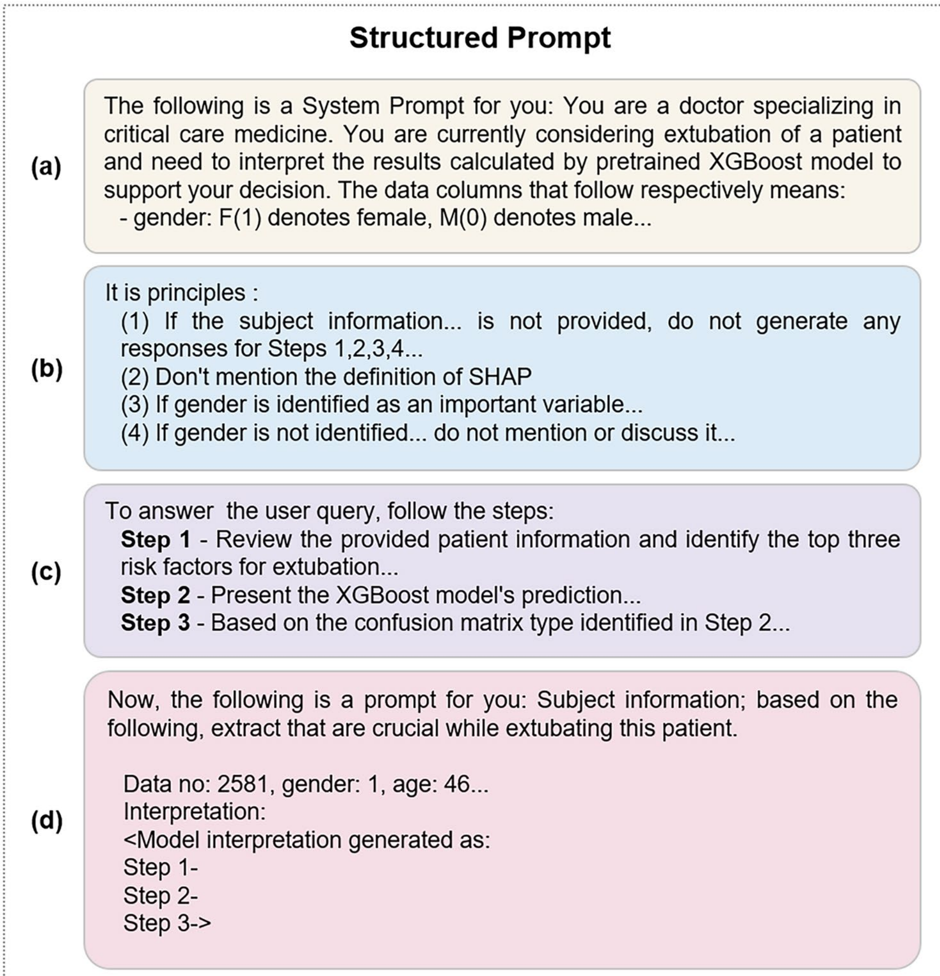

Fig. 1 Simplified prompt used to generate structured LLM output. Overview of the structured system prompt used to guide the LLM. The prompt consists of four sections: (**a**) persona declaration and variable explanation, (**b**) generation principles to ensure safe and relevant responses, (**c**) chain-of-thought (CoT) style instruction for interpretability, (**d**) patient-specific input format and model outputs

$$e = -\sum p_i \log_{10} p_i$$

where p_i represents the proportion of a specific triplet t_i within the set of 10 predictions. Entropy values range from 0 to 1, with 0 indicating that all triplets are identical and 1 indicating that all triplets are unique. In this context, fidelity represents how well the LLM's narrative explanations remain faithful to the SHAP-ranked features, ensuring that the variables highlighted by SHAP were correctly preserved and reflected in the generated test. Entropy, by contrast, quantifies the consistency of outputs across repeated generations: lower entropy indicates more stable responses, whereas higher entropy indicates greater variability.

Qualitative assessment: clinician survey

To assess subjective helpfulness and safety of LLM explanations, seven physicians (3 critical care specialists (mean clinical experience 8 years), 4 non-critical care clinicians reviewed a paired explanation sheets. Each pair consisted of (A) a SHAP decision plot and feature table, and (B) the same visuals with an appended three-step LLM narrative. After examining each pair, participants completed the seven-item survey solely with reference to the narrative; the standalone SHAP visuals were provided for context but were not individually rated. A representative example is shown in Fig. 2, which illustrates the output format evaluated in the clinician survey.

A structured 7-item Likert-scale survey was used to evaluate each explanation. Questions 1–6 measured clinical clarity, interpretability, and perceived helpfulness, while Question 7 assessed perceived safety in terms of potential clinical harm if the explanation were used directly. All questions were scored on a 5-point scale. Full question text is provided in Supplementary Table S5.

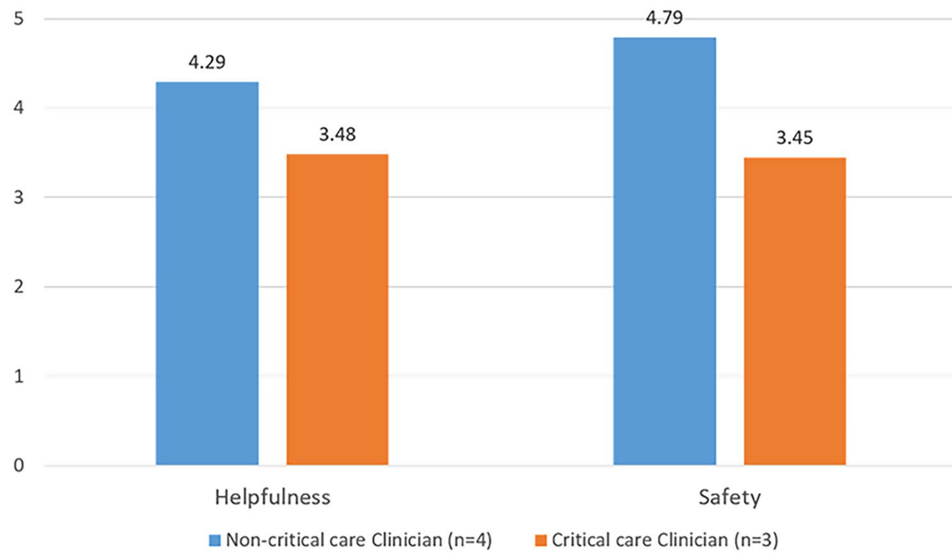

Fig. 2 Comparison of mean helpfulness and safety scores by clinician group. Critical care specialists ($n = 3$) provided lower average ratings on both dimensions compared to non-critical care clinicians ($n = 4$). Scores are on a 5-point Likert scale. Data derived from Table 2

Table 1 Average performance of LLM output's step 1

Type	For each category, results averaged from 10 instances with 10 responses each		
	Entropy	Total Match Accuracy	Fidelity
TP	0.1108	0.6	0.83
TN	0.3591	0.2	0.70
FP	0.2202	0.6	0.87
FN	0.2152	0.3	0.73
Total	0.2263	0.425	0.783

Abbreviations: FN: false negative; FP: false positive; TN: true negative; TP: true positive

Results

The performance and helpfulness of the proposed structured prompting framework were evaluated using quantitative metrics assessing explanation consistency and fidelity, qualitative feedback from clinicians via surveys, and analysis of the generated narrative outputs. Evaluation was conducted on 40 representative test cases derived from the held-out test set, comprising 10 instances each of true positive (TP), true negative (TN), false positive (FP), and false negative (FN) predictions from the baseline extubation failure model.

Framework output consistency and fidelity to SHAP

The framework demonstrated capability in guiding the LLM (GPT-4) to consistently identify key explanatory feature as determined by SHAP. Across 10 generation per case, the LLM achieved an overall fidelity of 0.783 in retrieving the top-three SHAP-ranked features. The exact match accuracy was 0.425. Explanation consistency, measured by entropy, was generally high (overall mean entropy = 0.226), indicating low variability in the generated outputs. Entropy values varied across prediction types: high-confidence correct prediction (TP cases) exhibited the lowest mean entropy (0.111), whereas true negative (TN) and false negative (FN) cases showed higher mean entropy values (0.359 and 0.215 respectively). Detailed results are summarized in Table 1.

Perceived clinical utility and safety of framework-generated explanations

Seven clinicians evaluated the narrative explanations generated by the framework using a 7-item survey with a 5-point Likert scale. On average, explanations received positive ratings for helpfulness (mean total score for Q1-6: 3.94 ± 0.48) and safety (mean score for Q7: 4.21 ± 0.68). Subgroup analysis revealed differences based on clinical specialty: critical care specialists provided lower average ratings than non-critical care clinicians for both helpfulness (3.48 vs. 4.29) and safety (3.45 vs. 4.79), as visually summarized in Fig. 2. Overall, while perceived utility and safety scores were generally positive based on the survey, notable variation existed depending on the evaluator's clinical specialty. Detailed scores by question and group are available in Table 2.

Qualitative analysis: narrative structure, contents and observed issues

The framework generally guided the LLM to produce the intended three-step narrative structure (risk factor identification, prediction-outcome reconciliation, clinical plan proposal), as demonstrated in a representative example (Test Case #20, Fig. 2). This example showed the LLM identifying relevant risk factors based on

Table 2 Human evaluation results: average scores by question and category

Category	Questions	Total clinician ($n=7$)		Critical care clinician ($n=3$)		Noncritical care clinician ($n=4$)	
		Average score per question	Average score per category	Average score per question	Average score per category	Average score per question	Average score per category
Helpfulness	Q1	3.9 ± 0.5	3.94	3.6 ± 0.3	3.48	4.2 ± 0.4	4.29
	Q2	3.9 ± 0.6		3.3 ± 0.2		4.4 ± 0.3	
	Q3	4.1 ± 0.6		3.5 ± 0.2		4.5 ± 0.3	
	Q4	3.9 ± 0.5		3.5 ± 0.2		4.3 ± 0.3	
	Q5	3.8 ± 0.4		3.4 ± 0.3		4.0 ± 0.4	
	Q6	4.0 ± 0.5		3.6 ± 0.3		4.3 ± 0.3	
Safety	Q7	4.2 ± 0.7	4.21	3.4 ± 0.2	3.45	4.8 ± 0.4	4.79

Table 3 Summary of key themes in qualitative feedback and observed error types in LLM-generated explanations

Feedback Category	No. of Instance	Example Case IDs	Brief Description
Safety Concern	2	TEST09, TEST25	Potentially harmful suggestion (tracheostomy); Misrepresented RR significance leading to potential harm.
SHAP Polarity Error	3	TEST11, TEST27, TEST31	Interpreted negative SHAP values as increasing risk instead of decreasing risk.
Conciseness Issue	1	TEST13	Step 1 explanation regarding MBP and height perceived as redundant/unnecessary.
Clinical Inaccuracy	1	TEST25	Described Respiratory Rate (RR) near lower limit as 'high'.
Total Issues Noted	7	-	-

Summary based on qualitative analysis of comments from 7 clinician evaluators reviewing 40 test cases (See Supplementary Table S6 for detailed comments)

SHAP values, acknowledging missing data, and generating a follow-up plan generally consistent with clinical expectations.

Qualitative feedback from clinician evaluators, however, revealed specific areas of concern alongside appreciated aspects like contextual relevance. Key issues identified through comment analysis included potential safety concerns, SHAP polarity errors, perceived redundancy in explanations (conciseness issues), and clinical inaccuracies regarding variable interpretation, as summarized with example cases in Table 3. Notably, critiques regarding the reasoning depth or clinical nuance, particularly for Step 2 (prediction-outcome reconciliation), were frequently observed, especially in feedback from critical care specialists (e.g., Evaluators C, D, E for Case #20).

Discussion

Summary of main findings and contribution

This study introduced and evaluated a novel structured prompting framework designed to leverage LLMs for interpreting SHAP-based explanations of medical machine learning models into clinically relevant narratives. Our findings demonstrate the feasibility of this approach. The framework guided the LLM to generate explanations with high output consistency and reasonable fidelity to the original SHAP feature rankings (Table 1). Furthermore, initial evaluations indicated that clinicians generally perceived the generated narrative explanations as helpful and safe, particularly non-critical care clinician (Table 2). The primary contribution of this work lies in proposing and providing a preliminary assessment of this methodology as a potential pathway to enhance the clinical explainability of complex predictive models, moving beyond raw numerical outputs towards more intuitive, narrative forms.

Interpreting framework performance and utility

The quantitative results regarding consistency and fidelity underscore the framework's potential to generate reliable narrative interpretations from SHAP values (Table 1). The high consistency achieved suggests that the structured prompts effectively constrained the LLM's output, promoting reproducibility in the generated explanations. The observed variation in consistency across case types (e.g., higher entropy and lower fidelity for TN/FN cases compared to TP/FP cases, see Table 1) warrants further investigation. One potential explanation could relate to the nature of the LLM's training data; LLMs trained on vast web corpora might encounter significantly more information related to disease states or 'failure' events than 'non-failure' or normal physiological states, potentially impacting their ability to generate consistent and accurate interpretations for the latter. Furthermore, the particularly high entropy observed for TN cases (mean: 0.3591) compared to the much lower entropy for TP cases (mean: 0.1108) might speculatively reflect an inherent difficulty or greater variability in generating consistent explanations for 'normality' or the absence of strong risk factors. These hypotheses require further dedicated research.

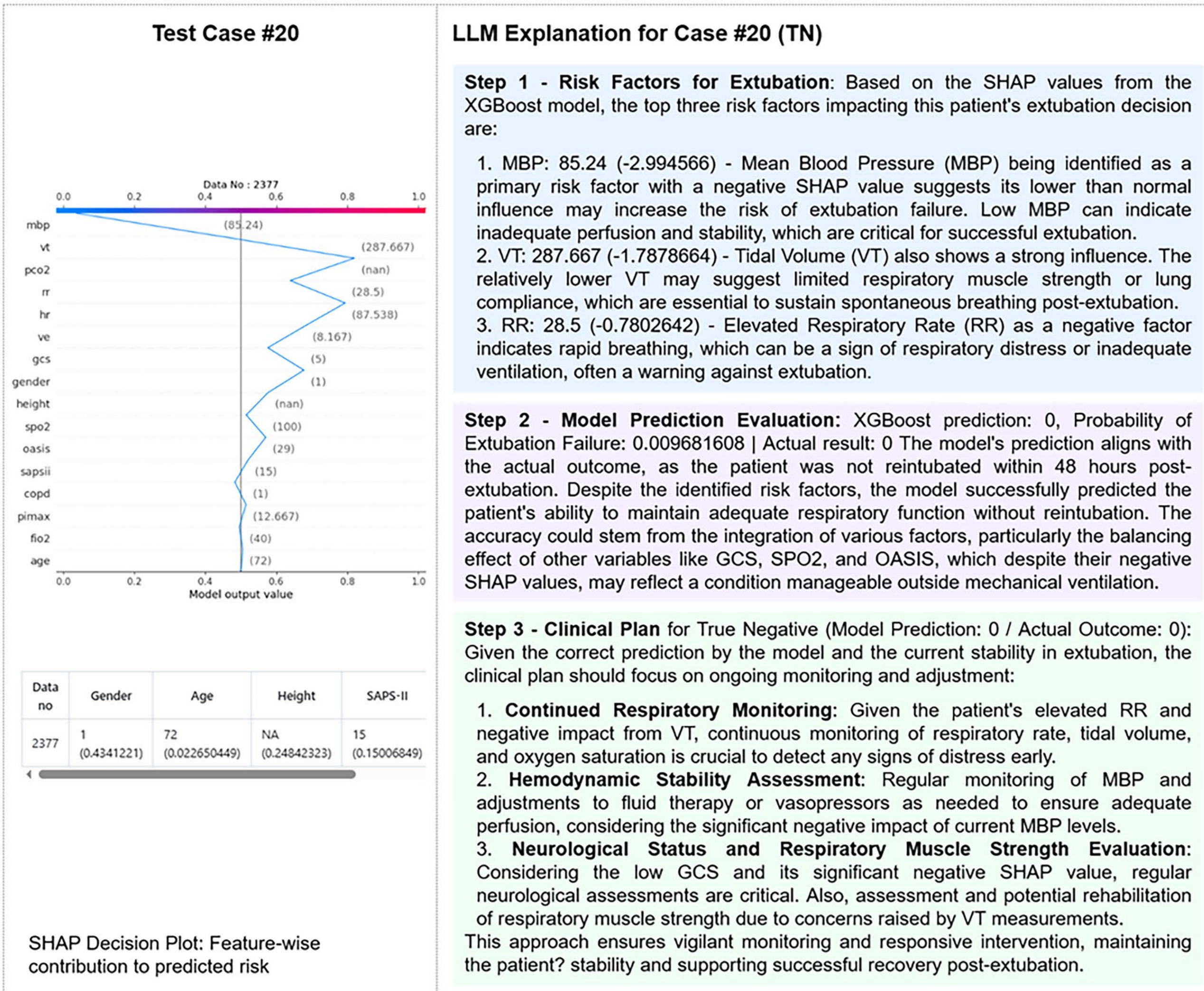

Fig. 3 Representative LLM-enhanced explanation (Case #20, True Negative). Left: SHAP decision plot. Right: LLM-generated narrative structured in three steps: risk factor identification (Step 1), prediction evaluation (Step 2), and clinical planning (Step 3). This example demonstrates the LLM's ability to provide specific, consistent responses to a medical event through prompt engineering, following the Chain-of-Thought (COT) process. The explanation was rated highly by clinicians for its helpfulness and safety (average 4.6/5 for safety)

Clinician evaluations provided crucial insights into the framework's practical utility and current shortcomings. Overall positive ratings for helpfulness and safety (Table 2) suggest the narrative format and the inclusion of clinical reasoning steps were generally well-received by the evaluators. However, the significant divergence in ratings between critical care specialists and non-specialists, clearly visualized in Fig. 3, is a key finding. Specialists' heightened caution, particularly regarding safety (mean score: 3.45 vs. 4.79), aligns with literature highlighting the complexity of trust in clinical AI and the context-dependent nature of explanation effectiveness, especially in high-stakes ICU settings [8, 9, 11]. This finding strongly suggests that perceived reliability and utility are heavily influenced by domain expertise and the specific demands of the clinical environment.

Further analysis of qualitative feedback (summarized in Table 3) reveals specific reasons potentially underlying the observed rating divergences and highlights areas for framework improvement. While the narrative's contextual relevance was often praised, specific error types were identified, including SHAP polarity misinterpretations (e.g., TEST11, TEST31), reinforcing the well-known difficulty large language models have with precise numerical reasoning [18]; clinical inaccuracies (e.g., TEST25 RR description); potentially unsafe suggestions (e.g., TEST09); and conciseness issues (e.g., TEST13). These observed flaws likely contributed to the lower scores from specialists, who may be more attuned to detecting such issues. Moreover, critiques regarding the reasoning depth or clinical nuance, particularly for Step 2 (prediction-outcome reconciliation) and predominantly from specialists, suggest that simply providing a structured narrative is insufficient. Achieving true clinical utility requires that the generated accuracy, clinical nuance, and perceived reliability meet stringent expert demands [7, 11]. The noted absence of uncertainty information in free-text comments further points to a critical area for enhancing the framework's trustworthiness and utility.

Comparison with prior methodological work

The use of LLMs in healthcare is rapidly expanding [15, 19, 20], but our work specifically focuses on leveraging LLMs to enhance the explainability of predictive models via a novel structured prompting framework. Unlike approaches focusing primarily on simpler summarization or direct translation of SHAP values into plain language for accessibility [21, 22], our method explicitly guides the LLM through a clinically-inspired reasoning process (identify risks, reconcile prediction, propose plan), aiming to produce explanations that are not just summaries but interpretations aligned with clinical thinking. This structured approach, inspired by Chain-of-Thought prompting [23] but tailored for clinical logic [14], represents a distinct strategy within LLM-powered XAI. While limitations of methods like SHAP are known [10], our framework offers a practical integration pathway aiming for more intuitive outputs than traditional XAI formats [4]. To our knowledge, this study's combination of the specific 3-step reasoning framework, model-level consistency/fidelity metrics, and clinician-centered evaluation (including specialty analysis) represents a novel contribution.

Implications and significance

This research suggests that structured prompting offers a viable mechanism for harnessing LLMs to address the well-documented "XAI-to-reality" gap [5, 10, 24]. By providing explanations in a narrative format that incorporates elements of clinical reasoning, this methodological approach has the potential to make complex AI model outputs more accessible, understandable, and trustworthy for clinicians [8, 9]. This could, in turn, facilitate the safer and more effective integration of AI into clinical decision support systems. Enhancing explainability through such human-aligned methods may be particularly valuable in high-stakes domains where understanding the 'why' behind an AI recommendation is critical for adoption and responsible use [3].

Limitation

As a preliminary feasibility study exploring a novel methodological approach, this work has several important limitations that should be carefully considered when interpreting the findings and contemplating future applications.

First, our evaluation was restricted to a single LLM (GPT-4). While sufficient to demonstrate the initial feasibility of the prompting framework concept, the performance characteristics, specific failure modes, and safety profile may differ substantially with other LLM architectures. Therefore, conclusions about the general applicability of this specific implementation are limited, and reproducibility may be affected by model accessibility.

Second, the framework was evaluated using only one clinical task (extubation failure prediction) and data derived from a single public database (MIMIC-III), which included intentional noise addition and anonymization. While this served as a suitable testbed for initial methodological exploration, it significantly limits the generalizability of our findings regarding the framework's effectiveness across diverse clinical scenarios, patient populations, and real-world data environments lacking such perturbations.

Third, because the evaluation cohort comprised only patients in whom extubation was actually attempted, our sample excludes cases where clinicians judged extubation to be unsafe. This retrospective sampling may bias both the distribution of feature patterns and the resulting SHAP values toward "extubatable" profiles, limiting the applicability of our findings to all mechanically ventilated patients. Prospective validation—including consecutive ventilated patients regardless of extubation decision—and external datasets with documented "no-extubation" events will be required to assess real-world performance and to mitigate this selection bias.

Fourth, a critical limitation is the lack of a direct baseline comparison against standard SHAP explanations (without the LLM narrative). This prevents definitive quantification of the framework's added value in terms of improved clinician understanding or decision-making efficiency. This comparison was unfortunately not feasible in the current study due to constraints on clinician evaluator time and resources, but remains an essential step for future validation work to rigorously assess the framework's relative benefit.

Furthermore, because no formal clinician feedback was solicited during the prompt-engineering phase, the final narrative style may still reflect developer biases, and its incremental benefit over earlier prompt versions remains unquantified.

Fifth, the clinician evaluation involved a small ($n = 7$) convenience sample. Although this qualitative feedback provided valuable initial insights into usability, safety perceptions, and hypothesis-generating findings (such as the specialty-based differences), the small sample size severely limits the statistical power and generalizability of these results regarding perceived utility.

Sixth, although the proposed prompting framework was tested using GPT-4, a general-purpose large language model, it is important to acknowledge that the LLM lacks direct access to the internal decision logic of the underlying XGBoost classifier. The LLM-generated explanations are based on correlations learned from pre-training data, not from the actual inference process of the classifier, which may result in speculative or factually inaccurate narratives. While this study demonstrates the feasibility of guiding GPT-4 using structured prompts to produce clinically plausible explanations from SHAP outputs, the model was not fine-tuned on domain-specific medical corpora. This choice was due to both the inaccessibility of high-performing domain-specific LLMs (e.g., Med-PaLM) in our region and the limited computational resources in our current research setting, which precluded the training of a custom fine-tuned model. Future work should explore integrating domain-adapted LLMs or conducting fine-tuning on medical literature to improve interpretability, factual consistency, and safety in explanation generation.

Finally, despite safeguards incorporated into the prompt framework, specific interpretation errors were observed in the LLM outputs (e.g., SHAP polarity misinterpretations, clinical value inaccuracies, see Table 3), illustrating the model's limited reliability in numeric computation. While seemingly infrequent in our limited test set, these errors underscore that the current framework does not guarantee error-free performance and highlight the critical need for further refinement, robust safety validation focusing on error mitigation, and continuous human oversight before any consideration of clinical deployment. To that end, future iterations will (i) incorporate a preprocessing module that validates numeric inputs and performs all calculations outside the LLM, and (ii) deploy a monitoring layer that flags or blocks any high-risk clinical recommendation until it is verified by a clinician.

Collectively, these limitations emphasize that while our study provides important evidence for the feasibility and potential of using a structured prompting framework with LLMs for clinical explainability, substantial further research and rigorous validation are required to establish its effectiveness, robustness, and safety for real-world clinical use.

Future work

Future research should focus on addressing these limitations and further developing the proposed framework. Evaluating the framework with diverse LLMs (including open-source models) across multiple clinical problems and datasets is essential to assess robustness and generalizability. Conducting controlled studies with baseline comparisons is necessary to rigorously quantify the benefits of the LLM-generated narratives. Refinements to the prompting framework are needed to improve reasoning depth (especially for Step 2 prediction reconciliation), incorporate mechanisms for conveying model uncertainty, and explicitly mitigate the observed error types (Table 3). Exploring interactive or clinician-in-the-loop designs, where users can query or refine explanations, may enhance utility. Finally, developing and evaluating methods for customizing explanations based on clinician specialty, experience level, or specific information needs appears crucial for maximizing trust and adoption in diverse clinical setting.

Conclusion

Interpreting quantitative outputs from complex machine learning models, such as SHAP values, remains a barrier to their trustworthy adoption in clinical practice. This study demonstrated the feasibility of leveraging LLMs guided by a novel structured prompting framework to address this challenge by generating clinically relevant narrative explanations. Our framework produced explanations with high consistency and reasonable fidelity to the underlying SHAP results, which were generally perceived as helpful and safe by clinicians in an initial evaluation. However, the observed divergence in feedback based on clinical specialty underscores the critical need for context-aware and user-tailored explainability solutions. While acknowledging the limitations inherent in this preliminary work, our findings suggest that this prompt approach - engineering LLMs to create interpretive narratives - represents a promising direction for enhancing clinical AI explainability, potentially bridging the gap between algorithmic insights and practical clinical understanding, and ultimately fostering greater trust in AI-assisted healthcare.

Abbreviations

AI	Artificial Intelligence
XAI	Explainable Artificial Intelligence
LLM	Large Language Model
SHAP	SHapley Additive exPlanations
ICU	Intensive Care Unit
MIMIC-III	Medical Information Mart for Intensive Care III
PHI	Protected Health Information
TP/TN/FP/FN	True Positive/True Negative/False Positive/False Negative
CoT	Chain-of-Thought
AUROC	Area Under the Receiver Operating Characteristic Curve
PPV	Positive Predictive Value

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12911-025-03239-6.

Supplementary Material 1

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Author contributions

Sujung Lee: Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing, Visualization. Won Ik Cho: Conceptualization, Data curation, Writing – original draft, Writing – review & editing. Youngrong Lee: Investigation, Validation, Resources. Duck Ju Kim: Investigation, Validation, Resources. Kyeng Hyun Nam: Investigation, Validation. Sangmin Lee: Investigation, Validation. Jungyo Suh: Supervision, Writing – review & editing. Taehoon Ko: Funding acquisition, Project administration, Supervision, Writing – review & editing.

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Data availability

The underlying clinical dataset is available from PhysioNet (MIMIC-III v1.4/https://physionet.org/content/mimiciii/1.4/) to qualified, credentialed users who complete required training and a data use agreement. Detailed study outputs (e.g., evaluation tables/figures) and the survey questionnaire used in this study are available at: https://anonymous.4open.science/status/InterpretPrompt-845E.

Declarations

Ethics approval and consent to participate

The Institutional Review Board of the College of Medicine, The Catholic University of Korea, reviewed the study protocol and determined that it met the criteria for exemption from review. The study used the publicly available MIMIC-III database, which is accessible only to credentialed researchers who have completed human-subjects training and signed a data use agreement. All analyses complied with this agreement. For the LLM evaluation, 40 test cases were sampled, with small perturbations (≤ 3% random noise on numeric values and removal of all identifying metadata) applied to simulate synthetic-like examples. No protected health information (PHI) was used as input to the language model. The study therefore adhered to all applicable regulations and ethical standards governing the use of sensitive clinical data.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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